Road Accident Data Analysis & Visualization

Interactive dashboard, risk simulator, location map and severity insights



Data Analytics Project Presentation

Problem Statement & Objectives



The problem

Road accidents cause loss of life and property every day. Raw accident records are spread across rows and columns, so patterns in time, place, weather and cause are hard to see.

Prepare the data

Clean, standardize and engineer features from raw records

Find patterns

Compare time, place, weather, cause and severity

Visualize insights

Dashboard, map and filters make results easy to explore

Estimate risk

Simulator shows an illustrative risk score for given conditions

Dataset Overview

Field	Type	Example
Accident_ID	Text	A-1042
Date, Time	Datetime	2024-03-14, 21:30
City	Category	Chennai
Latitude, Longitude	Float	13.08, 80.27
Weather	Category	Rainy
Road_Condition	Category	Wet
Cause	Category	Over-speeding
Severity	Category	Minor / Serious / Fatal
Casualties	Integer	2

9

field groups

Any public road accident CSV can be mapped to these fields.

Latitude and longitude power the map. Severity, Cause and City power the filters.

Project Workflow



Collect

Load accident records from a CSV dataset



Preprocess

Clean data, fix types and add time features



Explore

Find patterns with summaries and EDA



Visualize

Dashboard, map and severity charts



Interact

Filters, risk simulator and CSV download

Output: an interactive web app that turns raw rows into decisions

Data Preprocessing Steps

1. Load & inspect

Check shape, data types, summary statistics and null counts with info() and describe().

2. Missing values

Fill categories with the mode and numbers with the median. Drop rows without coordinates.

3. Remove duplicates

Drop repeated records using Accident_ID so each accident is counted once.

4. Types & features

Parse dates, then derive Hour, Weekday, Month and Time_of_Day (Day, Evening, Night).

5. Standardize labels

Trim spaces, fix casing and merge variants such as Rain, Rainy and Raining.

6. Validate & outliers

Keep coordinates in valid ranges, check casualties with IQR and encode severity levels.

Preprocessing in Python (Pandas)

```
import pandas as pd

df = pd.read_csv("accidents.csv")
df = df.drop_duplicates(subset="Accident_ID")

df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
t = pd.to_datetime(df["Time"], format="%H:%M", errors="coerce")
df["Hour"] = t.dt.hour
df["Weekday"] = df["Date"].dt.day_name()

df = df.dropna(subset=["Latitude", "Longitude"])
df["Weather"] = df["Weather"].fillna(df["Weather"].mode()[0])
df["City"] = df["City"].str.strip().str.title()
df["Severity"] = df["Severity"].str.title()

df.to_csv("accidents_clean.csv", index=False)
```

Why it matters

- Clean input means trustworthy charts
- Hour and Weekday drive the dashboard
- Valid coordinates make the map reliable
- One clean CSV is reused by every feature

Raw Data vs Clean Data

Raw data

- Duplicate accident records
- Mixed labels: Rain / rainy / RAINY
- Missing coordinates and weather
- Dates stored as plain text

Clean data

- One row per accident
- Consistent categories for filters
- Complete, map-ready coordinates
- Hour, Weekday and Month ready

Saved as accidents_clean.csv and loaded by the dashboard, map, simulator and filters.

Key Features

Accident Analysis Dashboard

Explore accident data through charts by hour, cause, city and weather.

Accident Risk Simulator

Change weather, road condition and time to see an illustrative risk score.

Accident Location Map

View dataset coordinates on a map with severity-coloured markers.

Severity Analysis

Compare minor, serious and fatal accidents across conditions.

Interactive Filters

Filter records by city, accident cause and severity.

CSV Download

Download the filtered or cleaned data as a CSV file.

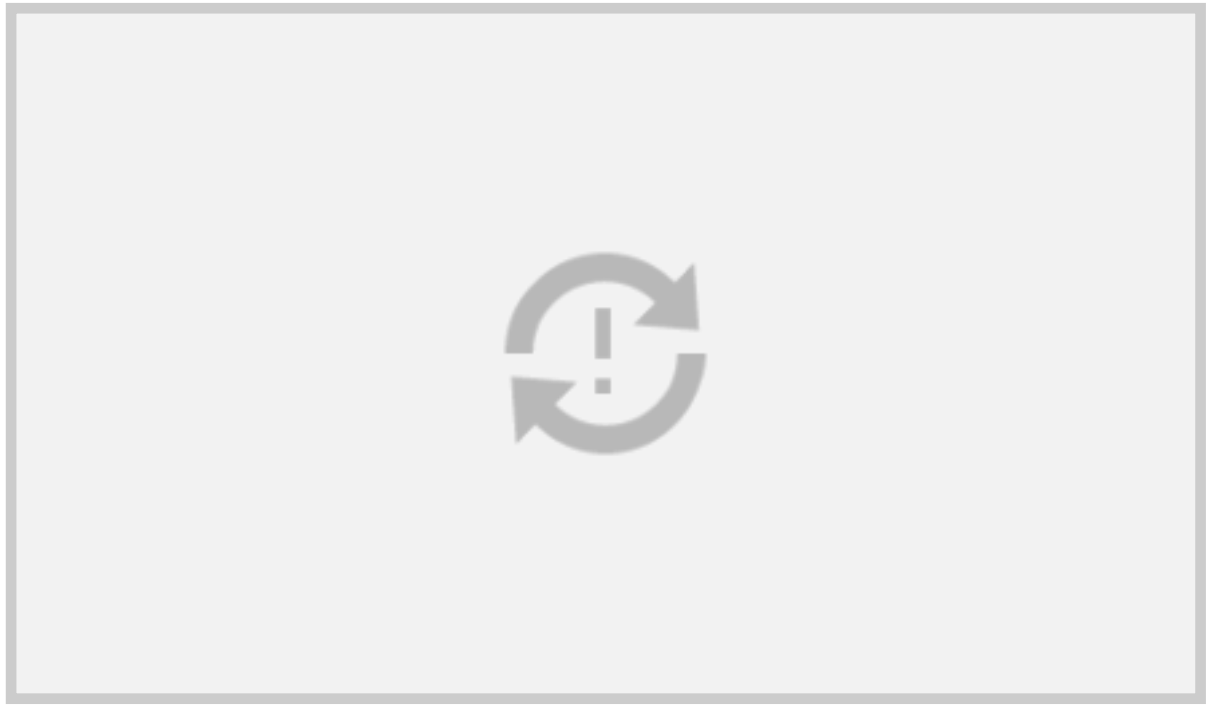
Accident Analysis Dashboard

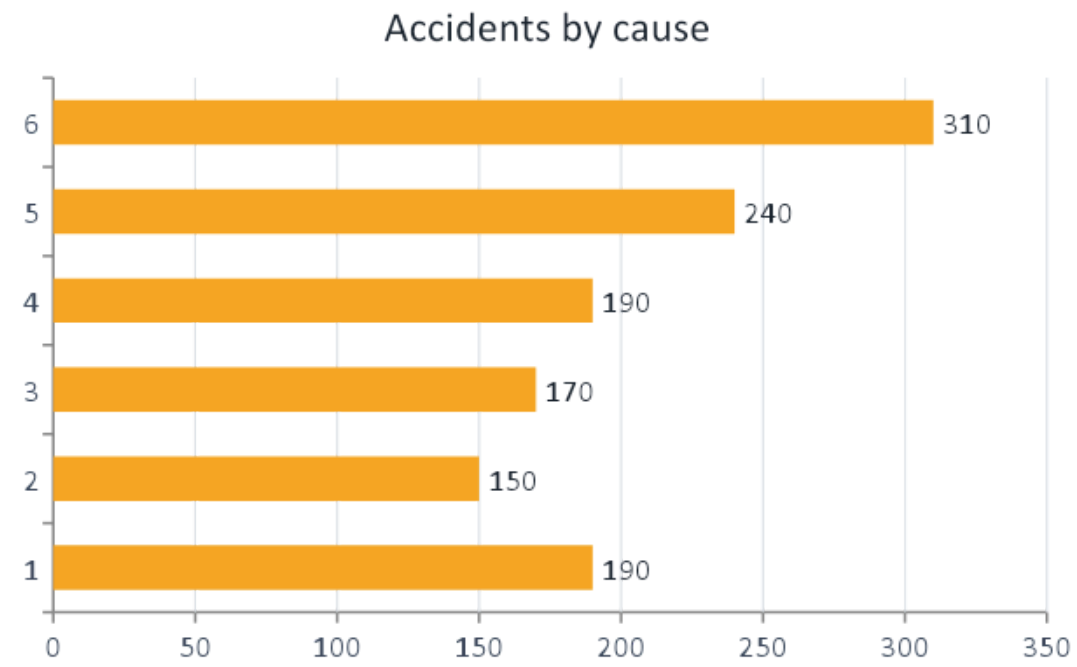
1,250
Total accidents

6-9 PM
Peak hours

Over-speeding
Top cause

Chennai
Highest accident city





Illustrative sample data

Accident Risk Simulator

Inputs

Weather

Rainy

Road condition

Wet

Time of day

10:00 PM (Night)

Risk score

67

out of 100

High risk

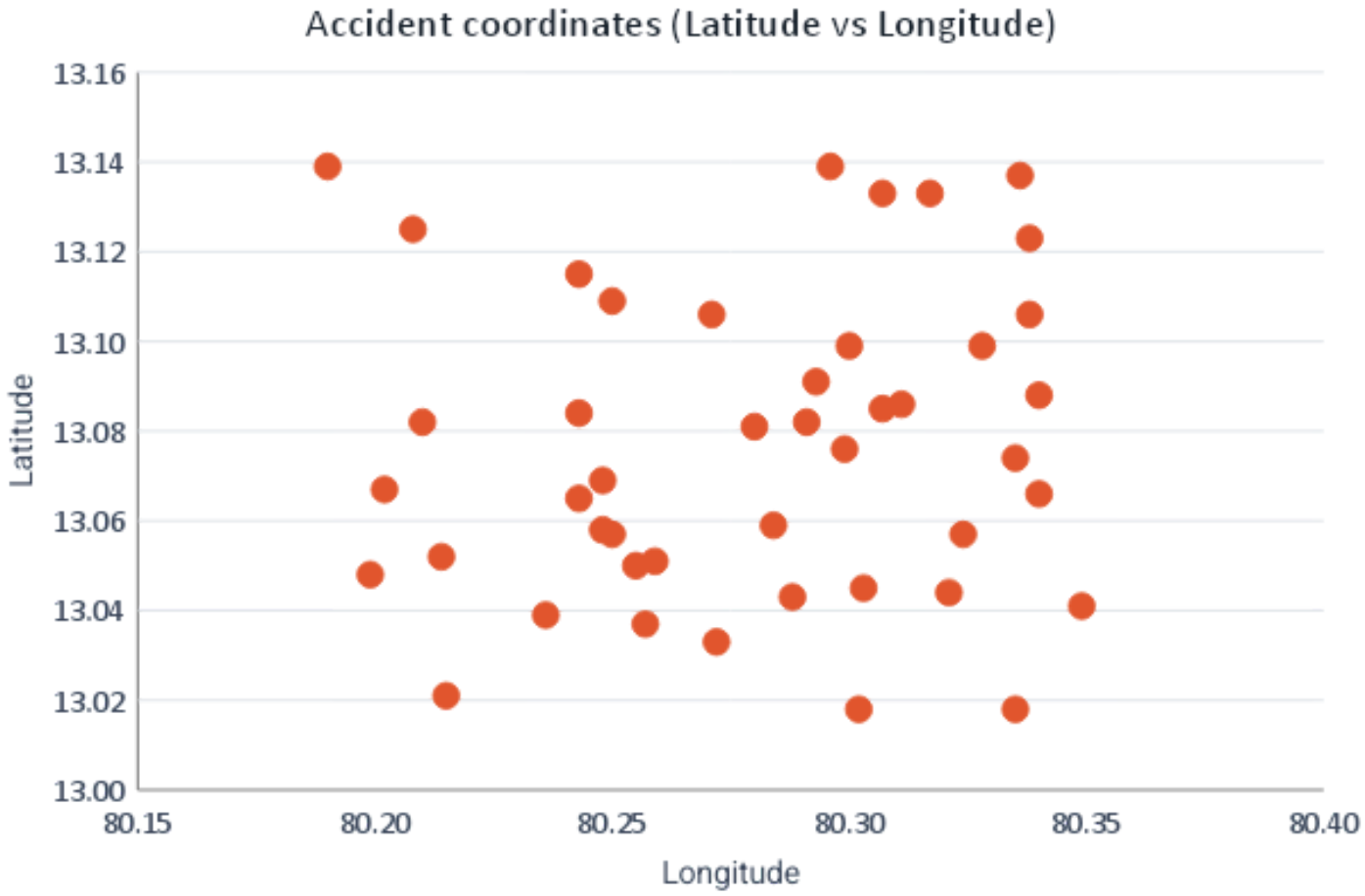
How the score is built

Factor	Points
Base score	20
Weather (Clear/Rain/Fog)	0 / +15 / +20
Road (Dry/Wet/Damaged)	0 / +12 / +18
Time (Day/Evening/Night)	0 / +10 / +20

Example: 20 + 15 + 12 + 20 = 67

Low below 35, Moderate 35-60, High above 60.
Illustrative score, not a real prediction.

Accident Location Map



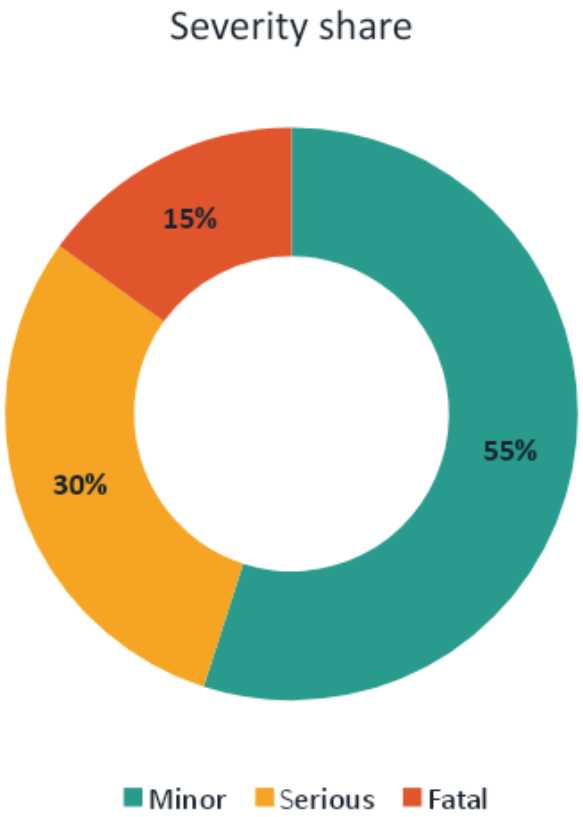
Sample coordinates around one city

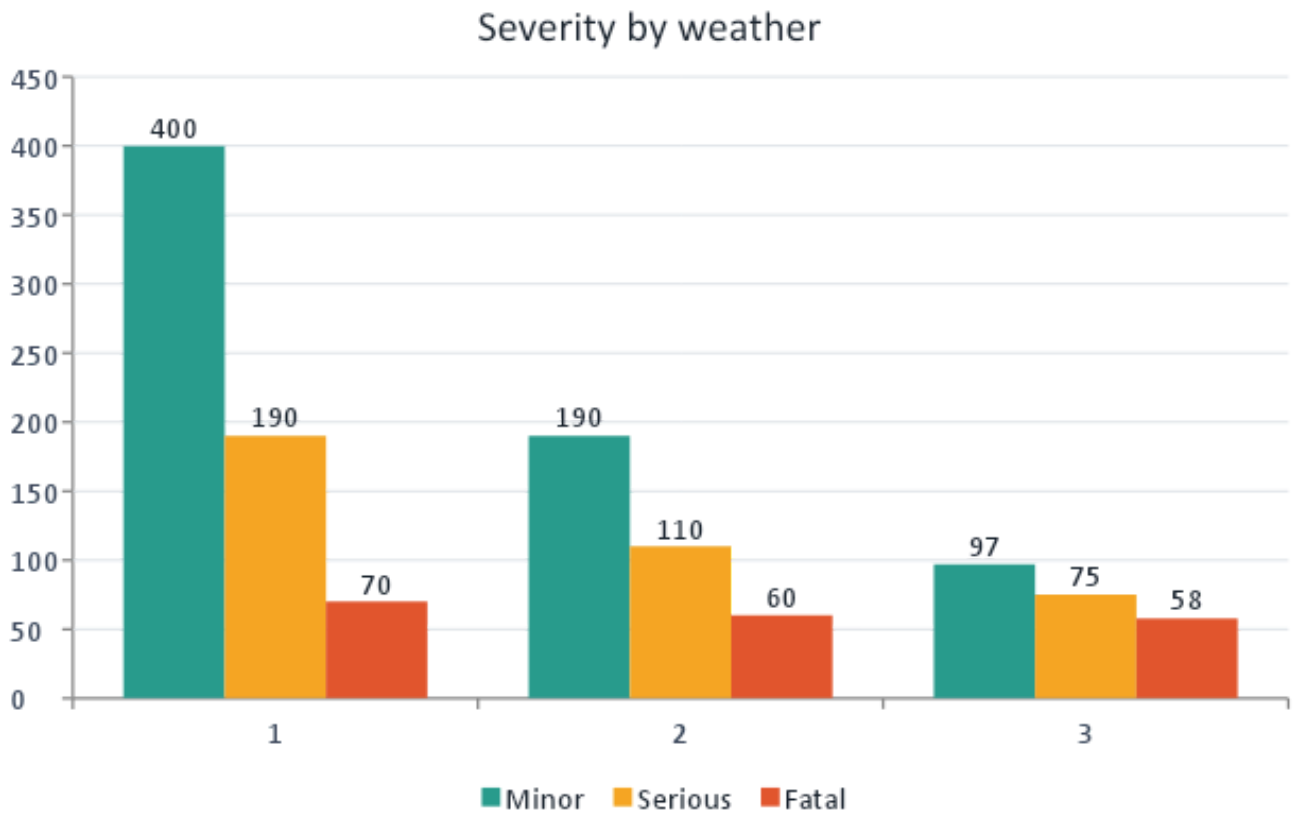
Map layers

- Markers coloured by severity
- Heatmap of accident hotspots
- Clusters when zoomed out
- Popup with date, cause and weather

Built with Folium or Plotly using the Latitude and Longitude columns.

Severity Analysis





Most accidents are minor, but the share of serious and fatal cases rises in rain and fog.

Interactive Filters & CSV Download

Interactive filters

City

Chennai | Mumbai | Delhi

Accident cause

Over-speeding | Distracted

Severity

Minor | **Serious** | **Fatal**

CSV download

City	Cause	Severity
Chennai	Over-speeding	Serious
Chennai	Over-speeding	Fatal
Chennai	Distracted	Serious

Download CSV

Downloads exactly the rows that match the current filters.

Suggested Tech Stack

Python	Core language
Pandas	Preprocessing and analysis
Plotly	Interactive charts
Folium	Accident location map
Streamlit	Web app, filters and download

Filters, map and CSV download (Streamlit)

```
sb = st.sidebar
city = sb.multiselect("City", df.City.unique())
cause = sb.multiselect("Cause", df.Cause.unique())
sev = sb.multiselect("Severity", df.Severity.unique())

view = df[df.City.isin(city) & df.Cause.isin(cause)
          & df.Severity.isin(sev)]

st.map(view.rename(columns={"Latitude": "lat",
                            "Longitude": "lon"}))
st.download_button("Download CSV",
    view.to_csv(index=False), "accidents.csv")
```

Conclusion & Future Scope

What we achieved

- Cleaned raw accident records with a repeatable pipeline
- Built dashboard, map and severity views
- Added filters, risk simulator and CSV export

Future scope

- Predict accident severity with machine learning
- Connect live data feeds or APIs
- Add alerts and a mobile-friendly view

Thank you. Questions?